



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Date of Examination: 25.04.2024

Session: (AN)

Duration: 3 hours

Full Marks: 100

Subject: No.: ME60310

Subject: Microfluidics

Department: Mechanical Engineering

This question paper booklet includes questions and accompanying contents that provide inputs, hints and outlines for answering the question. To demarcate between the question asked and the rest of the inputs given, the question asked is typeset in **BOLD** whereas the rest is in regular font. Further, for answering the questions, following common data can be used, if required: Avogadro constant (N_A) = $6.023 \times 10^{23} \text{ mol}^{-1}$, Boltzmann constant (k_B) = $1.38 \times 10^{-23} \text{ J K}^{-1}$, Proton charge (e) = $1.602 \times 10^{-19} \text{ C}$, Viscosity of an aqueous solution = 10^{-3} Pa. s , Density of water = 10^{-3} kg/m^3 , Permittivity of water = $80 \times 8.854 \times 10^{-14} \text{ F/m}$. All questions are compulsory.

Q1. If particles are included in a fluid, there may be different ways of accounting for their contribution in the theory of fluid flow, either directly or indirectly, based on certain approximations. One such model system is known as micropolar fluid, where the rotational contribution of the included particles is cumulatively quantified by the micro-rotation vector (ω), in addition to the standard translational velocity vector ($\mathbf{V}$). For an incompressible micropolar fluid, the continuity, linear momentum conservation and angular momentum conservation equations respectively read: $\nabla \cdot \mathbf{V} = 0$; $\rho \frac{D\mathbf{V}}{Dt} = -\nabla p - (\mu + \kappa) \nabla \times \nabla \times \mathbf{V} + \kappa \nabla \times \omega$; $\rho \frac{D\omega}{Dt} = (\alpha + \eta + \gamma) \nabla(\nabla \cdot \omega) - \gamma \nabla \times \nabla \times \omega + \kappa \nabla \times \mathbf{V} - 2\kappa \omega$, where p is the pressure; μ, κ are known as the coefficients of shear and vortex viscosities, respectively; α, η, γ are known as the coupled viscosities.

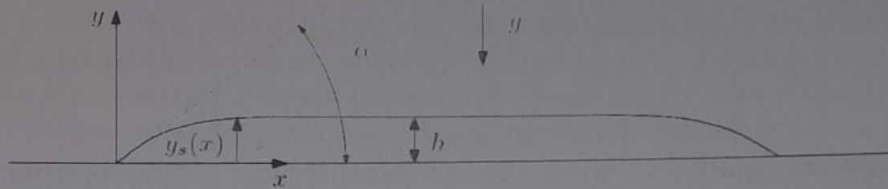
Consider the low Reynolds number flow of a micropolar fluid, described by the above governing equations, taking place in a parallel plate microchannel of height H . The flow may be considered as fully developed, so that: $\mathbf{V} = (u(y), 0, 0)$ and $\omega = (0, 0, \omega(y))$.

(a) **Non-dimensionalize the governing equations by using the length scale $\rightarrow \sqrt{\frac{\gamma}{\kappa}}$, velocity scale $\rightarrow U_c$ (a characteristic flow velocity, depending on the how the flow is actuated), micro-rotation scale $\rightarrow U/L_c$, and pressure gradient scale $\rightarrow \frac{U(\mu + \kappa)}{L_c}$. From analyzing the same, find an expression for U_c when the flow is driven by a pressure gradient dp/dx , when $H \gg \sqrt{\frac{\gamma}{\kappa}}$. Does the length of the microchannel have to play a role to play in this estimation?**

(b) Consider the following boundary conditions: $u=0$ at the walls, $\frac{du}{dy}=0$ at the centerline; $\omega=0$ at the centerline, $\omega|_w = \beta \frac{d\omega}{dy}|_w$ at the walls, where β is the non-dimensional coefficient known as the micro-rotation slip length and subscript w denotes the walls. **Based on these boundary conditions, derive expressions for the velocity and the micro-rotation.**

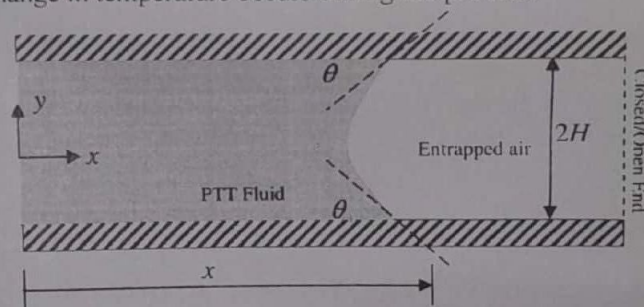
[15 marks]

Q2. A liquid of density ρ and liquid-gas surface tension γ has been spilled on a horizontal plate so that it forms a very large puddle whose depth (in the central parts) is h , as shown in the figure below. Consider the region near the edge of the puddle, which can be viewed to good approximation as two-dimensional. If the contact angle is α , **derive an expression for the shape of the liquid surface $y_s(x)$.**



[15 marks]

Q3. Consider a parallel-plate microchannel having height $2H$ and length L , as shown in the figure below, having one closed end and one open end through which a filling liquid (single-stranded DNA solution) is introduced for bio-analysis. This channel is initially occupied by air but is gradually filled up by the said liquid, which is not a perfectly viscous but also has partial elastic features as the DNA strands are something like stretchable springs connected by beads. Altogether, this is equivalent to a viscoelastic fluid, having partly viscous and partly elastic characteristics. The fluid has density ρ , whereas the surface tension at the fluid-gas interface is γ_s . The equilibrium contact angle at the fluid-solid-air interface as it fills up the channel is θ , which does not change as the capillary front progresses. Consider the air entrapped at the closed end to be an ideal gas for which no significant change in temperature occurs during the process.



Over the regime of capillary filling dynamics considered here, the flow is approximated as fully developed. As against a Newtonian fluid, the stress and strain rates for the viscoelastic fluid are not proportionately related, but coupled as (following the PTT constitutive model): $f(\tau_{kk})\tau_{xx} = 2\lambda u_{,y}\tau_{xy}$, $f(\tau_{kk})\tau_{yy} = 0$,

$f(\tau_{kk})\tau_{xy} = \eta u_{,y} + \eta\tau_{yy}u_{,y}$. Here, $\tau_{kk} = \tau_{xx} + \tau_{yy}$, and the subscript $,y$ denotes derivative with respect to y .

(a) **Derive an expression for the fully developed velocity profile of the DNA solution.**

(b) **Find an expression for the average flow velocity, based on the same.**

(c) **Find an expression for the viscous drag force on the fluid. State any approximations made.**

(d) Considering the fluid as a lumped body, the net force acting on the same can be described as:

$$\frac{d}{dt}\left(2\rho Hx \frac{dx}{dt}\right) = F_{st} - F_{visc} - F_{grav} - F_{gas}$$

Here, F_{st} , F_{visc} , F_{grav} and F_{gas} respectively denote surface tension force (the driving force), viscous resistance, gravitational force and retarding force exerted by the entrapped gas in a closed capillary. **Write the expressions of these forces as applicable for this specific model.**

(e) Assume p_i as the pressure of the entrapped gas and p_0 as the atmospheric pressure. Using Boyle's law:

$$2p_i(L-x)H = 2p_0LH$$

Taking this into account, **derive a governing differential equation that describes the dynamic displacement of the liquid-gas front as a function of time. Considering suitable length scales and time scales of the problem, derive a non-dimensional form of the same equation, and identify the**

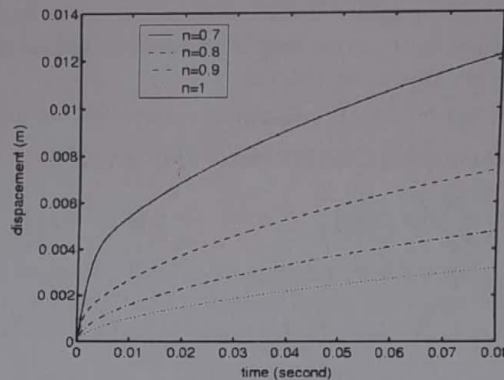
relevant non-dimensional parameters influencing its solution alongside their physical significance. There is no need to solve the equation.

(f) Derive a limiting-case of the above equation when the capillary is having both ends open. There is no need to solve the equation. [15 marks]

Q4. In a real-life healthcare application, there is interest in understanding how blood fills up a microchannel, dynamically. The physical situation is as follows. The blood sample first gets induced into the channel. As it progresses, the capillary front displacement of blood depends on a competition of the surface tension and viscous forces. For estimating the viscous force, the following constitutive model is used: The shear stress

$$\tau_{xy} = k \left(\frac{du}{dy} \right)^n, \quad \text{where the value of } n \text{ is typically between } 0.7 \text{ to } 1. \text{ The contact angle varies dynamically as}$$

the blood progresses inside the capillary. By solving the reduced order model equation similar to the one described in the previous question, an analyzer obtains the characteristics of displacement versus time of the blood sample for different values of n , as shown in the graph below.



Provide an explanation of the nature of the graphs in this figure, for different values of n .

[5 marks]

Q5. Consider a parallel plate microchannel of height $2H = 10 \mu\text{m}$ containing solution of NaCl in water at a concentration of $n_0 = 1 \mu\text{M}$. The viscosity (η) and permittivity (ϵ) correspond to that of a standard aqueous solution. An electric field of the form: $E = E_0 \sin(\Omega t)$ acts along the channel axis, as applied externally, where $E_0 = 100 \text{ V/cm}$ and $\Omega = 1 \text{ Hz}$.

(a) Under what physical conditions will this electric field be able to induce a flow of the solution along the channel axis? If so, will it necessarily be unidirectional?

(b) In an ionized state of the salt solution, the concentration of each ionic species, n_i , satisfies the following

charge conservation equation: $\frac{\partial n_i}{\partial t} + \nabla \cdot \vec{j}_i = 0$, where the ionic flux is given by: $\vec{j}_i = \underbrace{n_i \mathbf{V}}_{\text{Advection}} + \underbrace{-b_i n_i \nabla \mu_i^M}_{\text{Diffusion}}$.

Here μ_i^M is the molar electrochemical potential of the ion, which is the sum of the chemical potential and the electrical potential (ψ), and b_i is the ionic mobility. From pure dimensional considerations, find the unit of the ionic mobility and make an estimate of the same in terms of the ionic diffusivity D_i and the friction factor f (defined as the drag force on the ion per unit relative velocity with respect to the solution). Hence, derive a distribution of n_i in terms of the electrical potential along the transverse (y) direction. For deriving your expression, assume that, at the channel centerline: $\psi = \psi_0$, $n_i = n_i^0$, and, at the no-slip plane:

4f ω (freq. of oscillation) is high, the adjustment effect can't

$\psi = \psi_0 = 20$ mV. State the assumptions that you make for your analysis. Can the values of ψ_0 and n_i^0 be specified from the given problem data? Justify.

(c) Write a mathematical expression in differential equation form that physically represents the fact that the total charge accumulated in the fluid over a surface is nothing but the volumetric charge density integrated over the total volume of the fluid. Using the same, and the expressions for n_i as derived in part

(b) above, find an expression for the free charge density distribution in the fluid for each ion type, in its most simplified form.

(d) Consider the following dimensionless parameters: $\chi = \frac{y}{\lambda}$ where $\frac{1}{\lambda} = \sqrt{\frac{2n_0 e^2}{\epsilon k_B T}}$, $\kappa = \left(\frac{\Omega \chi^2}{\nu}\right)^{\frac{1}{2}}$, and

$U = \frac{u}{u_{HS}}$, where $u_{HS} = -\epsilon \psi_0 E_0 / \eta$. Using these, derive the dimensionless form of the governing

differential equation for fluid flow in its most simplified form. Mention the physical implications the different dimensionless parameters appearing therein.

(e) Considering $U(\chi, \theta) = \text{Im}[\exp(i\theta) F(\chi)]$, obtain a solution for the above equation in terms of the particular integral and the complementary function. It is given that, for the differential equation:

$$\frac{d^2 F}{d\chi^2} - i\kappa^2 F(\chi) = \frac{1}{2n_0 \alpha} \left(n_+^h \exp\left(\frac{ze}{k_B T}(\psi_h - \psi)\right) - n_-^h \exp\left(-\frac{ze}{k_B T}(\psi_h - \psi)\right) \right),$$

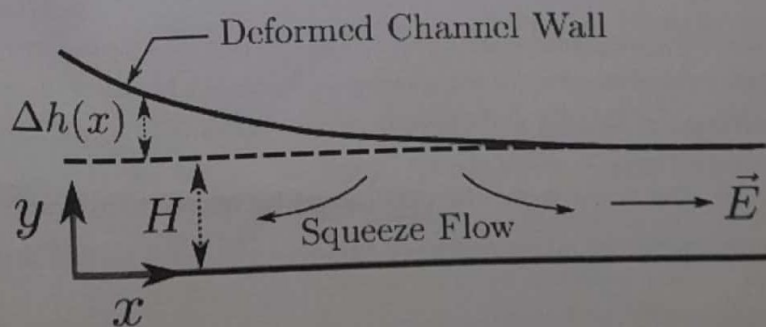
, the particular integral is given by, $PI =$:

$$\begin{aligned} & \frac{1}{2\kappa\alpha\sqrt{i}} \left[\exp(-\sqrt{i}\kappa\chi) \int_0^\chi \exp(\sqrt{i}\kappa\chi) \left(-\frac{n_+^h}{2n_0} \exp\left(\frac{ze}{k_B T}(\psi_h - \psi)\right) + \frac{n_-^h}{2n_0} \exp\left(-\frac{ze}{k_B T}(\psi_h - \psi)\right) \right) d\chi \right] \\ & - \frac{1}{2\kappa\alpha\sqrt{i}} \left[\exp(\sqrt{i}\kappa\chi) \int_0^\chi \exp(-\sqrt{i}\kappa\chi) \left(-\frac{n_+^h}{2n_0} \exp\left(\frac{ze}{k_B T}(\psi_h - \psi)\right) + \frac{n_-^h}{2n_0} \exp\left(-\frac{ze}{k_B T}(\psi_h - \psi)\right) \right) d\chi \right] \end{aligned}$$

(f) If the value of n_0 is changed to 1 mM, find an expression of the flow velocity in its most simplified form.

[18 marks]

Q6. Consider a parallel plate microchannel of height H as in the figure below, where the bottom wall is rigid and the top wall is deformable (see the figure below). Both the walls develop the same surface charge due to adsorption of ions from the bulk fluid, so that the potential at the shear plane is given by $\psi = \psi_0$.



Under the application of an axial electric field E and axial pressure gradient $\frac{dp}{dx}$, the top wall deforms slowly.

The stress distribution within the same is governed by the following equation: $\frac{\partial \sigma_{ij}}{\partial x_j} = 0$, where

$$\sigma = G \left(\frac{\partial s_i}{\partial x_j} + \frac{\partial s_j}{\partial x_i} \right) + \lambda \frac{\partial s_k}{\partial x_k}, \text{ with } \mathbf{s} \text{ being the displacement field, } G \text{ and } \lambda \text{ being the Lamé constants for the}$$

deformable wall material, assumed here to be homogeneous, isotropic and linearly elastic. The velocity of flow and displacement of the wall are coupled by the boundary conditions of stress balance at their interface.

(a) What is physically the 'shear plane' described in the above?

(b) Assuming that the wall-deformation is small and taking place dominantly along the y-direction, derive the above boundary condition in its most simplified form. Hence, derive an expression for s_y as a function of p , $\Delta h(x)$, and the elastic modulus: $E_m = 2G + \lambda$, where $\Delta h(x) = h(x) - H$.

(c) Extracting the pressure and hence the pressure gradient from the above analysis, derive an expression for the velocity field in terms of h . What is the implication of 'slowly' deforming wall in your derivation for this?

(d) Write a differential equation that physically represents the incompressibility of the flow. Integrating that equation across the depth of the deformable channel, derive a governing differential equation that describes the variation of h as a function of x and time, t .

(e) Assuming that ionic conductivity does not vary along x , we have $\frac{d}{dx}[E(x)h] = 0$. Using this and the preceding derivations, derive a governing differential equation for Δh as a function of x and t .

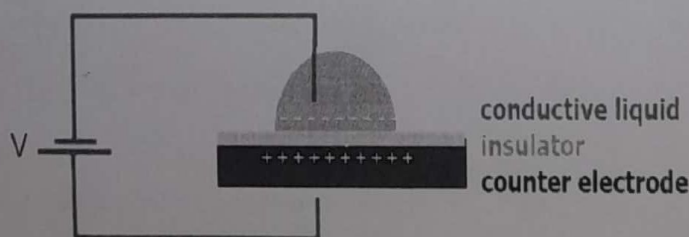
[17 marks]

Q7. Consider a droplet of a conductive liquid sitting on an insulator film which is atop a substrate having embedded electrodes. When an electric field is applied across the same, charges accumulate at the droplet-insulator interface, like a capacitor. The free energy at this interface now has an electrical contribution because of this charge accumulation, so that $E_{el} = -\frac{1}{2} \int \mathbf{E}(\mathbf{r}) \cdot \mathbf{D}(\mathbf{r}) dV$, where $\mathbf{E}(\mathbf{r})$ and $\mathbf{D}(\mathbf{r})$ are the electric field and

the electrical displacement at a position vector $\mathbf{r}$ and dV is an elemental volume. In this equation, $\mathbf{D}(\mathbf{r})$ is given by: $\mathbf{D}(\mathbf{r}) = \epsilon(\mathbf{r})\mathbf{E}(\mathbf{r})$, where $\epsilon(\mathbf{r})$ is the permittivity of the medium at $\mathbf{r}$. For evaluating the above integral, the liquid is considered to be a perfect conductor. Further, the electrostatic energy can be split into two parts. One part arises from a parallel plate capacitor formed by the droplet and the electrode, with $C = cA_{sl}$, where $c = \epsilon_d/d$, d being the thickness of the dielectric layer and ϵ_d being the corresponding permittivity. The second part arises from the stray capacitance along the edge of the droplet, which is neglected.

(a) Based on these assumptions, obtain an expression for E_{el} in its most simplified form. Justify the assumptions and explain how they are used for your derivation.

(b) Using the above result, derive an expression for the difference in solid-liquid interfacial energy with and without electric field effects.



④ If ω (freq. of oscillation) is high, the adjustment $\rightarrow$ effect can't

(c) Write an expression of the contact angle from the force balance at the interface (tangential component) in terms of the surface energies. Hence find an expression for the difference in the cosine of the contact angle with and without electric field. Does this difference depend on the solid-liquid interfacial energy?

(d) The phenomenon described as above can alternatively be perceived as an outcome of an extra stress (electrical stress) in addition to the hydrodynamic stress. The electrical stress can be described as:

$$\tau_{ij} = \epsilon \left(E_i E_j - \frac{1}{2} \delta_{ij} E_k E_k \right). \text{ To evaluate the resultant electromechanical force, note that the tangential}$$

component of the electrical field at the surface of the liquid droplet vanishes (why?), and accordingly, one may express the components of E on a plane with direction cosines n_i as $E_i = E n_i$. Utilizing the Cauchy's theorem relating the traction vector with the stress tensor components, one can write: $F_i = \int \tau_{ij} n_j dA$. Assuming a

uniform distribution of E over the droplet surface, derive an expression for the magnitude of $\frac{F_i}{A}$ (commonly known as electrocapillary pressure, p_{ec}), in terms of the equivalent change in the surface tension, $\Delta\gamma$, with and without electric field effects, the dielectric layer thickness, dielectric constant and the voltage applied.

(e) Consider the flow of a continuous liquid slug sandwiched in a channel of height H , for which the momentum equation reads: $\mu \frac{d^2 u}{dy^2} = \frac{dp}{dx}$, where y represents the transverse direction to the flow. The difference in inside

and outside pressure across an axial span of Δx can be described as: $\Delta p = \frac{2\Delta\gamma}{H}$, where $\Delta\gamma$ is the corresponding surface energy differential. Dividing the above expression by Δx and taking the limit as the same tends to zero, obtain an expression for $\frac{dp}{dx}$ in terms of the local surface tension gradient. Using the boundary

conditions: $u = 0$ at $y = 0$ (channel surface) and $du/dy = 0$ at $y = H/2$ (channel centreline), and assuming $\left\langle \frac{d\gamma}{dx} \right\rangle$ as the space averaged value of the surface tension gradient over the slug interface, find an expression of the velocity profile in the slug in terms of $\left\langle \frac{d\gamma}{dx} \right\rangle$. Hence find an expression for the average velocity.

(f) Consider now that this change in surface tension is due to the electric field effect described as above, and the voltage variation across the length of the slug can be described as: $V = \bar{\phi} - \langle \phi \rangle - \phi_0$, where $\bar{\phi}$ is the arithmetic mean of the electrical potentials at the two ends of the slug of length L (i.e., $\bar{\phi} = \frac{\phi_0 + \phi_L}{2}$, the subscript '0' being an indicator of the receding meniscus and the subscript 'L' being an indicator of the advancing meniscus of the slug), $\langle \phi \rangle = \frac{1}{L} \int_0^L \phi dx$, and $\phi_0 = Q/c$, Q being the charge (per unit area) present at the interface in absence of the applied potential, ϕ (Q includes both externally applied charges and charges due to ion adsorption from the electrolyte). Further, assume $\frac{dV}{dx} \approx \frac{\phi_L - \phi_0}{L}$ (i.e., a linear variation of the potential), and $\bar{\phi} \approx \langle \phi \rangle$. Considering $Q \approx 5 \mu C/cm^2$, $\phi_L - \phi_0 \approx 1 V$ and $H/L \approx 0.1$, estimate the order of magnitude of the mean velocity of the slug.

[15 marks]